Advanced Programming 2025

Testing Informational Efficiency in NBA Moneyline Betting Markets

Final Project Report

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Abstract

This project examines whether NBA moneyline betting markets exhibit exploitable inefficiencies using machine learning techniques. With the use of fifteen seasons of historical game and betting odds data, the study aims to test whether improved predictive accuracy can translate into statistically significant or economically meaningful, consistent betting profits, thereby providing evidence of market inefficiency.

The research develops advanced predictive modeling framework using engineered features that capture market expectations and game context, including bookmaker odds, point spreads, totals, historical team performance, and additional relevant variables. Three classification models—Logistic Regression, Random Forest, and Gradient Boosting—are evaluated using a season-based train–test split to avoid look-ahead bias. Model performance is assessed through standard classification metrics as well as expected value–based profitability measures.

The results indicate that Logistic Regression obtains the strongest out-of-sample predictive performance, demonstrating moderate discriminatory power and reasonably well-calibrated probability estimates. Nevertheless, expected value analysis and rank based significance tests conducted on out-of-sample test set reveal no statistically significant or meaningful relationship between model-predicted edge and realized returns.

The main contribution of this paper is an empirical demonstration that, despite improvements in predictive accuracy, machine learning models fail to reveal persistent inefficiencies in NBA betting markets when applied to publicly available information. These findings support the hypothesis of efficient markets in the context of sports betting markets.

Keywords: data science, Python, machine learning, market efficiency, NBA

Contents

1 Introduction

Sports betting markets are often viewed as environments where informed individuals believe they can outperform bookmakers by recognizing patterns or applying personal knowledge. From an economic point of view, however, this intuition is questionable. If profitable strategies were widely accessible, bookmakers would adjust their pricing accordingly, and any systematic advantage would disappear. For a market to function efficiently, constant mispricings should be rare or extremely difficult to detect. Logically, bookmakers have strong incentives to price the games accurately, as false odds would lead to their loss.

This conflict between human overconfidence and economic reasoning is what motivated me to do this project. Using fifteen seasons of NBA data, the goal is to evaluate whether moneyline betting markets display signs of inefficiency that a statistical or machine learning approach could exploit. The project combines two areas of interest: basketball analytics and empirical testing and econometrics.

To address this question, the analysis focuses on an advanced machine learning framework that incorporates engineered features capturing market expectations and game context, such as point spreads, totals, home-court effects, and historical team performance. Preliminary baseline specifications were explored to illustrate how much information is already embedded in bookmaker odds but are omitted for brevity, as they do not add any meaningful context to the conclusions. Several classification algorithms are evaluated using a season-based train–test split, and the best-performing model is selected based on out-of-sample predictive performance. The selected model's probability estimates are then used to evaluate expected value and test for potential market inefficiencies.

The main research question is therefore: Are NBA moneyline betting markets informationally efficient, or can statistical and machine learning models identify pricing inefficiencies that translate into positive expected value?

The remainder of this report is organized as follows. Section 2 presents the research question and reviews relevant literature on market efficiency and sports betting. Section 3 describes the dataset and outlines the methodological framework, including feature engineering, modeling approaches, and evaluation metrics. Section 4 discusses key implementation choices and technical considerations. Section 5 addresses code organization and reproducibility. Section 6 presents the empirical results, including predictive performance, expected value analysis, and statistical significance testing. Section 7 concludes by summarizing the findings, discussing limitations, and highlighting directions for future research.

2 Literature Review / Related Work

2.1 Market Efficiency in Sports Betting

The concept of market efficiency originates from financial economics, most notably the Efficient Market Hypothesis (EMH) introduced by Fama (1970). Under EMH, asset prices fully reflect all available information[1], implying that it is not possible to consistently achieve abnormal returns using publicly available data. While the Efficient Market Hypothesis provides a useful benchmark, its empirical validity has been widely debated[2]. This debate makes examining market efficiency in settings outside traditional financial markets, such as sports betting, more attractive. Although sports betting markets differ from traditional financial markets, the similarities are evident: prices (odds) are set by professional market makers (bookmakers in this case) updated frequently, and influenced by both information and participant behavior.

Prior research generally finds that major sports betting markets are close to efficient, particularly in popular sports with high liquidity. Levitt (2004) shows that bookmakers do not necessarily aim to set odds equal to true probabilities, but instead choose prices that balance betting volume and manage risk exposure[3]. As a result, betting odds reflect a combination of true outcome probabilities and strategic pricing, making any potential inefficiencies unexploitable.

Several empirical studies that examine football, basketball, and other major sports report that, while small biases may exist, these tend to disappear once transaction costs and bookmaker margins are taken into account. Overall, the literature suggests that mature betting markets incorporate publicly available information rapidly, leaving limited opportunities for systematic arbitrage.

2.2 Machine Learning for Sports Outcome Prediction

Advances in machine learning have led to increased interest in applying predictive models to sports outcomes. Common approaches include logistic regression, random forests, gradient boosting, and neural networks. These methods have been shown to outperform simple heuristic or baseline models in terms of classification accuracy and ranking performance in complex and high-dimensional settings[4].

In the context of sports betting, machine learning models typically use features derived from historical team performance, market odds, game context, and other publicly available information. Tree-based methods, such as random forests and gradient boosting, are particularly popular due to their ability to capture nonlinear relationships and feature interactions. Logistic regression, while simpler, remains widely used because of its interpretability and strong generalization properties in settings with limited signal-to-noise ratios.

Although many studies show improvements in predictive accuracy when machine learning methods are applied to sports data, the magnitude of these improvements is often not impressive. Moreover, higher predictive accuracy does not necessarily imply that a model can generate profitable betting strategies[5], especially in markets where information asymmetry is present.

2.3 Prediction Accuracy versus Betting Profitability

A key distinction emphasized in the literature is the difference between predicting outcomes accurately and achieving positive expected value in betting markets. Metrics such as accuracy or ROC AUC measure how well a model ranks outcomes but do not account for market prices. Profitability depends on whether predicted probabilities differ meaningfully from probabilities implemented by the bookmaker after accounting for margins and risk management techniques. In other words, even if the model can calibrate the probabilities better than the market, it will not necessarily lead to direct profits.

Several studies highlight that models capable of predicting winners better than the market may still fail to generate positive expected value once evaluated out of sample.[6] This discontinuity arises because bookmakers incorporate predictive information into odds and adjust prices dynamically in response to betting flows. As a result, improvements in probability estimation alone are often insufficient to overcome market efficiency.

2.4 Positioning of This Project

Building on this literature, the present project evaluates NBA moneyline betting markets using a machine learning framework that emphasizes out-of-sample evaluation and expected value analysis. Rather than focusing on predictive accuracy alone, the analysis explicitly tests whether improved probability estimates translate into statistically significant or economically meaningful betting profits. By combining predictive performance metrics with expected value and rank-based significance tests on a true out-of-sample dataset, the project aims to provide a rigorous assessment of market efficiency in NBA betting markets.

3 Methodology

3.1 Data Description

The dataset used in this project consists of NBA regular-season games spanning fifteen seasons, from the 2008–2009 season through the 2022–2023 season. This dataset includes variables like team identities, final outcomes, and betting-related variables such as moneyline odds, point spreads, and game totals. The data were obtained from publicly available sports statistics and betting odds sources that aggregate historical NBA results and bookmaker prices.

In total, the dataset contains over 18,000 games, providing a large sample size suitable for statistical analysis and machine learning modeling. Each observation corresponds to a single game from the perspective of one team, resulting in a binary outcome variable indicating whether the team won or lost the game.

The feature set includes both market-based and contextual variables. Market-based features include bookmaker's probabilities derived from moneyline odds, point spreads, and totals. Contextual features capture game-specific information such as home-court advantage and historical team performance measures constructed using only information available prior to each game. These historical features are computed in a rolling manner to avoid the use of future information.

The quality of the data is good overall, with certain issues including missing values for certain betting variables in early seasons as well as occasional inconsistencies across data sources. These are removed during preprocessing to ensure consistency. Generally speaking, the dataset is well-suited for studying market efficiency due to its size, coverage, and relevance to a highly liquid betting market.

3.2 Approach

3.3 Approach

From a data science perspective, this project integrates data collection, feature engineering, predictive modeling, and evaluation within a single empirical pipeline. Rather than focusing solely on classification accuracy, the analysis combines statistical learning methods with economic evaluation criteria to assess whether predictive improvements translate into consistent profits.

The technical approach combines supervised classification models with a framework based on expected value evaluation. The goal is not only to predict game outcomes but also to assess whether the probabilities attributed by the model differ meaningfully from probabilities derived from bookmaker's odds in a way that could generate profitable betting opportunities.

Three classification algorithms are considered: Logistic Regression, Random Forest, and Gradient Boosting. Logistic Regression serves as a strong baseline due to its interpretability and stability, while the two tree-based methods are included to capture potential nonlinear relationships and feature interactions. All models are trained to output predicted probabilities rather than hard classifications, as probability estimates are required for expected value calculations.

Formally, Logistic Regression models the probability of a team winning a game as

$$P(Y = 1 \mid X) = \frac{1}{1 + \exp\left(-(\beta_0 + \beta^\top X)\right)},$$

where $Y \in \{0, 1\}$ denotes the game outcome, X is the vector of input features, and β represents the model parameters estimated from the training data. In this project, Logistic Regression is used purely for probabilistic prediction rather than causal inference.

Prior to modeling, preprocessing is performed. The data are cleaned to address missing observations and inconsistencies, bookmaker-implied probabilities are obtained by transforming moneyline odds into corresponding probabilities, and historical team performance features are constructed using only information from games played before each matchup. Where necessary, features are scaled to ensure numerical stability, particularly for models that are sensitive to differences in feature magnitude.

Model performance is evaluated using standard classification metrics, including accuracy, ROC AUC, log loss, and the Brier score. ROC AUC is used to assess the ability of the models to correctly rank winning and losing outcomes across probability thresholds. Log loss evaluates the quality of probabilistic predictions by penalizing overconfident incorrect forecasts, while the Brier score measures the calibration of predicted probabilities relative to observed outcomes. Together, these metrics provide a comprehensive evaluation of both discriminatory power and probability reliability, which is essential for expected value analysis. Model selection is based primarily on out-of-sample ROC AUC.

To assess market efficiency, probabilities assigned by the model are compared to bookmaker's probabilities to compute an edge predicted by the model. Expected value (EV) is then calculated for each game based on the predicted probability and the corresponding payout. Games are grouped into EV buckets to evaluate whether higher predicted edge corresponds to higher realized returns.

Given a predicted win probability $\hat{p}$ and corresponding bookmaker odds with decimal payout o, the expected value (EV) of a unit bet is computed as

$$\text{EV} = \hat{p} \cdot (o - 1) - (1 - \hat{p}),$$

where $\hat{p}$ is the model-predicted win probability and o denotes the decimal odds. Games are grouped into EV buckets to evaluate whether higher predicted edge corresponds to higher realized returns.

3.4 Implementation

All analysis is implemented in Python using standard data science libraries, including pandas and NumPy for data manipulation and scikit-learn for model training and evaluation. The codebase follows a modular structure that separates data preprocessing, modeling, and evaluation to ensure clarity and reproducibility.

To illustrate the implementation of the modeling pipeline, the following code snippet shows how the Logistic Regression model is trained and used to generate predicted probabilities on the test set.

```python
from sklearn.linear_model import LogisticRegression

# Initialize and train the model
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

```

```
# Generate predicted probabilities on the test set
y_prob = model.predict_proba(X_test)[:, 1]
```

Listing 1: Logistic Regression training and probability prediction

This example demonstrates the core modeling logic used throughout the project. Predicted probabilities produced by the models serve as inputs for both performance evaluation metrics and expected value calculations.

4 Results

4.1 Experimental Setup

All the analysis was performed using Python. Model training and evaluation were performed using the scikit-learn library. To ensure reproducibility, fixed random seeds were used where applicable.

A season-based train–test split was employed to avoid look-ahead bias. Games from the 2008–2009 through 2020–2021 seasons were used for training, while the 2022–2023 seasons were reserved as a true out-of-sample test set. This setup mirrors real-world betting conditions, where only past information is available at the time of prediction.

Three classification models were trained and compared: Logistic Regression, Random Forest, and Gradient Boosting. Hyperparameters were chosen conservatively to balance predictive performance and model stability, with particular care taken to avoid overfitting given the noisy nature of sports outcome data.

4.1.1 Predictive Performance Evaluation

Performance Metrics Predictive performance is assessed using accuracy, ROC AUC, log loss, and the Brier score. Accuracy measures the proportion of correctly predicted outcomes but does not account for prediction confidence. ROC AUC evaluates a model's ability to correctly rank winning and losing outcomes across probability thresholds and is therefore used as the primary model comparison criterion. Log loss penalizes overconfident incorrect predictions, while the Brier score measures the calibration of predicted probabilities.

Table 4.1 reports the out-of-sample predictive performance of the three models on the 2022–2023 test set.

Model	ROC AUC	Log Loss	Brier Score
Logistic Regression	0.719	0.621	0.210
Random Forest	0.690	0.652	0.229
Gradient Boosting	0.711	0.634	0.218

Table 1: *Out-of-sample predictive performance on the 2022–2023 test set*

Logistic Regression achieves the highest ROC AUC, indicating the strongest overall ranking performance. Gradient Boosting performs comparably, while Random Forest exhibits weaker generalization performance, consistent with increased sensitivity to noise in a low signal-to-noise setting.

Model Selection Given its superior out-of-sample ROC AUC and favorable calibration metrics, Logistic Regression is selected as the primary model for subsequent expected value and market efficiency analysis. In addition to its competitive performance, Logistic Regression offers greater interpretability and stability relative to more complex tree-based models.

4.1.2 Expected Value Analysis

To assess whether improved predictive performance translates into betting profitability, model-implied probabilities are compared to bookmaker-implied probabilities to compute expected value (EV) for each game. Expected value is calculated assuming a unit stake and reflects the average profit or loss implied by the model's predicted probability and the bookmaker's payout.

Games in the out-of-sample test set are grouped into buckets based on model-predicted edge. Average realized returns are then computed for each bucket to examine whether higher predicted edge corresponds to higher realized returns.

The results show that average EV remains negative across most buckets, reflecting the bookmaker margin. While higher-confidence buckets tend to exhibit less negative average EV than lower-confidence buckets, the pattern is noisy and unstable. Occasional near-zero or slightly positive average EV outcomes appear in higher buckets, but these are not consistent across buckets or time periods

4.1.3 Statistical Significance Testing

To formally assess whether the observed relationship between model-predicted edge and realized returns is statistically meaningful, a Spearman rank-order correlation test is conducted. Spearman's correlation coefficient measures the strength of a monotonic relationship between two variables based on their ranks and does not assume linearity or normality. It is defined as

$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)},$$

,where d_i denotes the difference between the ranks of the two variables for observation i, and n is the number of observations.

The null hypothesis tested is

$$H_0 : \rho = 0,$$

which corresponds to the absence of a monotonic relationship between model-predicted expected value and realized returns. On the out-of-sample test set, the estimated Spearman correlation is $\rho = 0.60$, with a p-value of 0.067. As this p-value exceeds the conventional 5% significance threshold, the null hypothesis cannot be rejected at standard confidence levels.

4.2 Visualizations

To complement the numerical results, several visual diagnostics are presented using the out-of-sample test set. These figures help illustrate model discrimination, probability calibration, and the relationship between predicted edge and realized betting outcomes.

Figure 1 displays the ROC curve for the selected Logistic Regression model. The curve lies consistently above the diagonal reference line, indicating that the model is able to discriminate between wins and losses better than random guessing. This visual evidence is consistent with the reported test-set ROC AUC and confirms the presence of moderate predictive power in a highly competitive market.

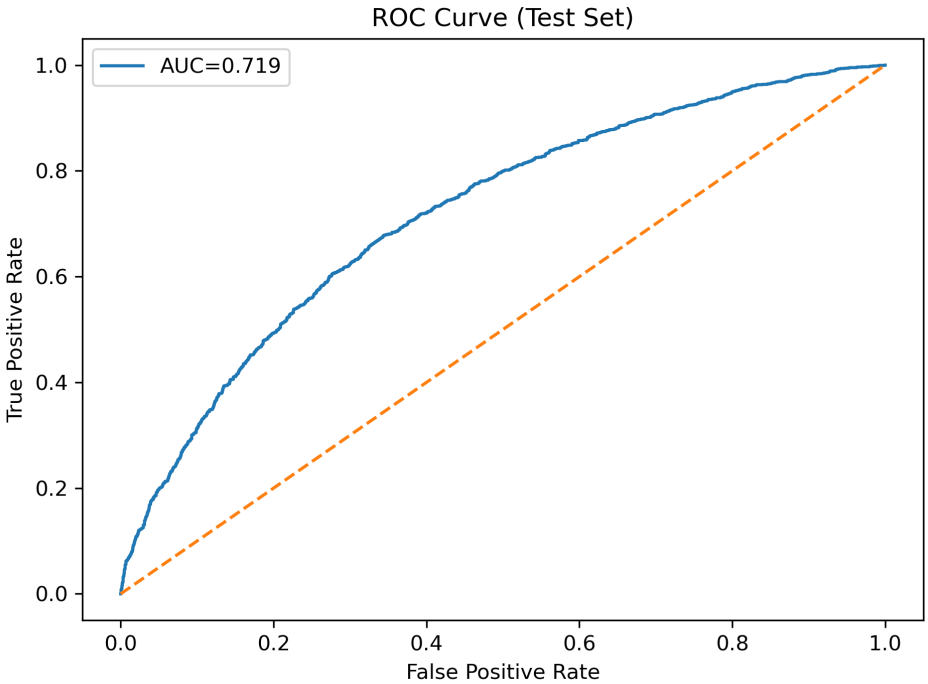

Figure 1: *ROC curve for the Logistic Regression model on the out-of-sample test set.*

Figure 2 presents the calibration curve for the test set. Predicted probabilities closely track observed win frequencies across probability bins, suggesting that the model's probability estimates are reasonably well calibrated. This property is crucial for expected value analysis, as miscalibrated probabilities would lead to unreliable estimates of betting profitability.

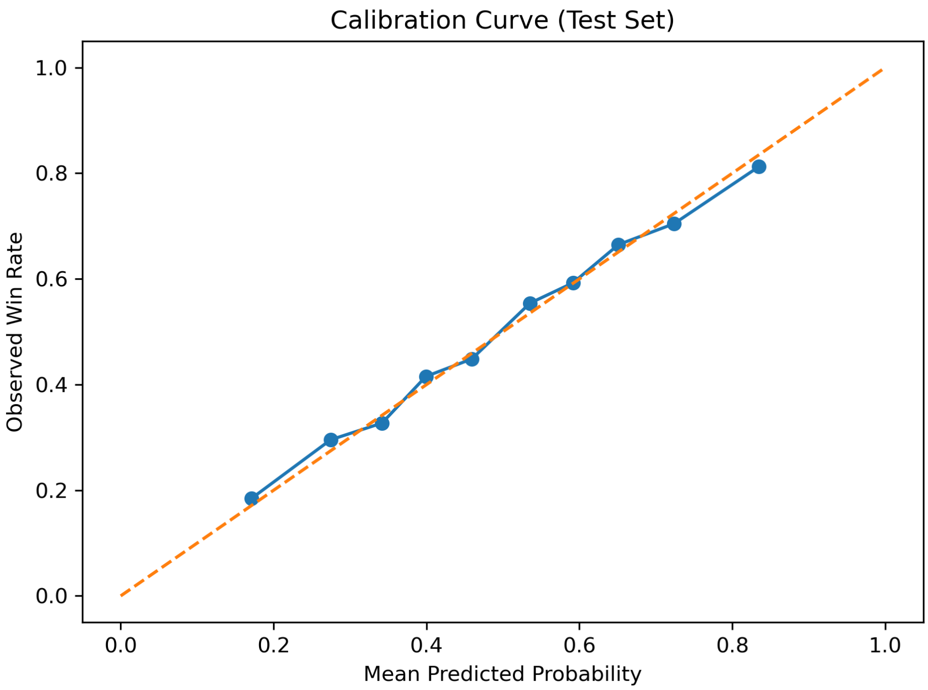

Figure 2: *Calibration curve for predicted probabilities on the out-of-sample test set.*

Figure 3 reports average realized returns by predicted edge bucket. While higher-confidence buckets tend to exhibit less negative average returns, the overall pattern remains noisy and unstable. Average returns are negative across most buckets, reflecting the bookmaker margin, and no clear monotonic relationship emerges out of sample.

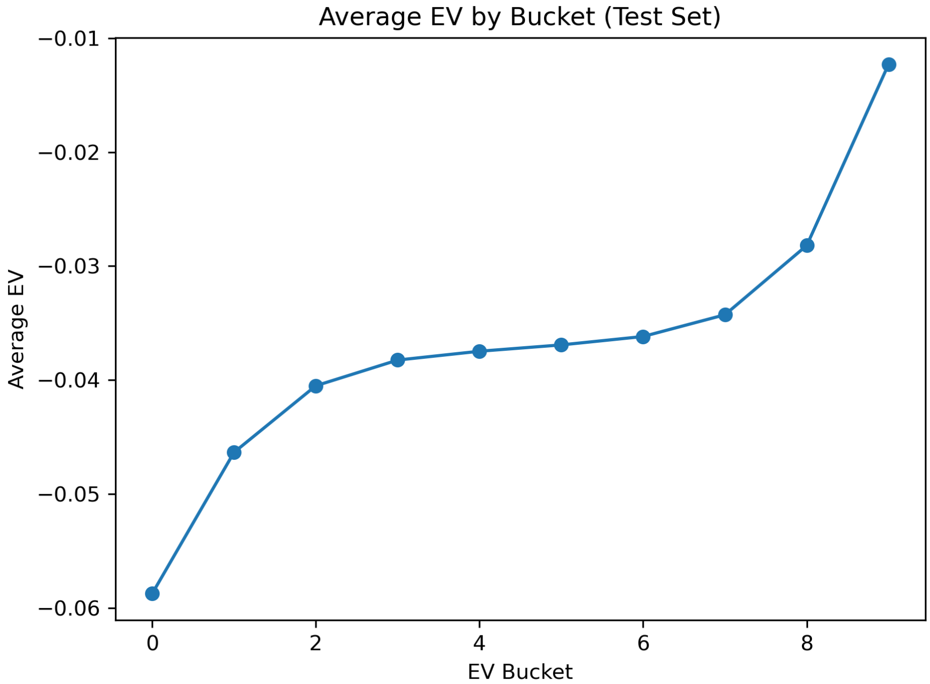

Figure 3: *Average realized return by predicted edge bucket on the out-of-sample test set.*

Finally, Figure 4 illustrates the relationship between predicted edge and realized return at the individual game level. The wide dispersion of outcomes highlights the substantial variance inherent in sports betting and helps explain why apparent predictive advantages do not translate into consistent profitability.

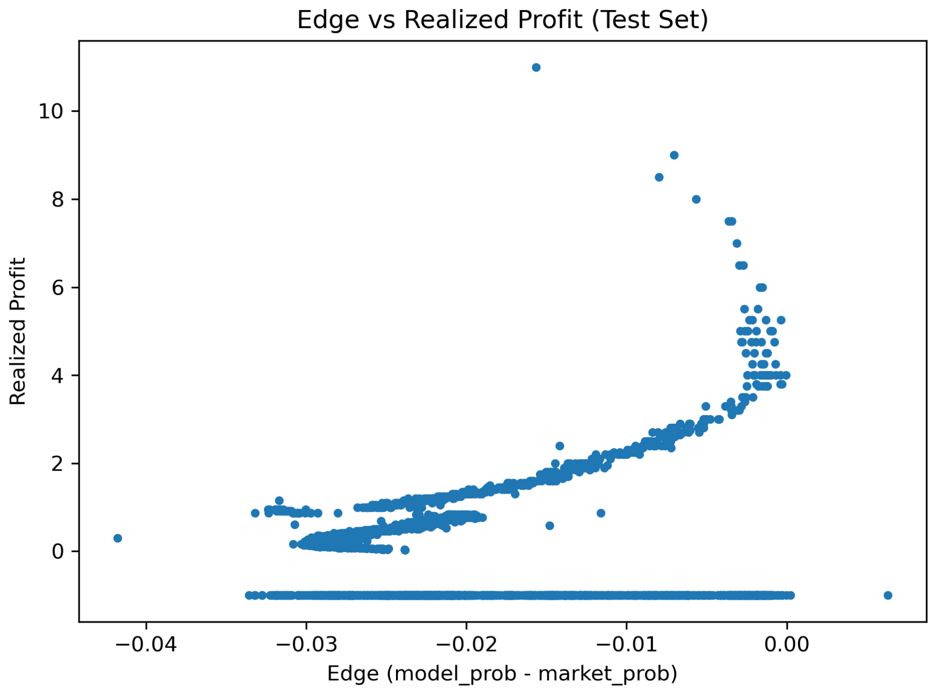

Figure 4: *Predicted edge versus realized return at the game level on the out-of-sample test set.*

5 Discussion

Overall, the modeling framework performs reasonably well from a predictive standpoint. Among the evaluated models, Logistic Regression achieves the strongest out-of-sample performance, with a test-set ROC AUC of approximately 0.72. This indicates that the model is able to distinguish between wins and losses better than random guessing, despite the inherent uncertainty of NBA game outcomes. Probability calibration diagnostics further suggest that predicted probabilities are reasonably aligned with observed outcome frequencies, which is a necessary condition for meaningful expected value analysis.

However, improvements in predictive accuracy do not translate into economically meaningful betting profits. Expected value analysis on the out-of-sample test set shows that average EV remains negative across most confidence buckets, reflecting the bookmaker margin. Although higher-confidence buckets exhibit slightly improved average EV relative to lower-confidence buckets, the pattern is noisy and unstable. Statistical significance testing using a Spearman rank-order correlation yields a moderate correlation between predicted edge and realized returns, but the relationship is not statistically significant at conventional levels. This suggests that any apparent ordering is likely driven by variance rather than a persistent betting edge, which is an important distinction emphasized in the literature on machine learning and asset pricing[7].

Several challenges help explain these findings. First, NBA moneyline betting markets are highly liquid and closely monitored, allowing sportsbooks to rapidly incorporate publicly available information into odds. Second, my models rely exclusively on historical and publicly available data, which limits their ability to capture real-time information such as injuries, lineup changes, or last-minute breaking news. Finally, the NBA games' outcomes are subject to high randomness, as any team could easily win on a given night and even the biggest underdogs often perform a lot better than expected. This implies that even well-calibrated probability estimates will not lead to consistent profits with certainty, especially over short evaluation periods

Overall, the results align with theoretical expectations from market efficiency. While machine learning models can improve probability estimation, they do not appear sufficient on their own to overcome the informational and structural advantages held by bookmakers in betting markets. This is further pointed out in recent work on the application of advanced artificial intelligence which states that even highly sophisticated models face structural limits in competitive environments rich in information, reinforcing the difficulty of achieving persistent economic gains[8].

6 Conclusion and Future Work

6.1 Summary

This project investigated whether NBA moneyline betting markets present exploitable inefficiencies using fifteen seasons of historical game and odds data. An advanced machine learning pipeline was developed to predict game outcomes and generate model-implied probabilities, which were then evaluated using both predictive performance metrics and expected value–based tests on a true out-of-sample dataset.

The results show that while Logistic Regression achieves moderate predictive accuracy and produces reasonably calibrated probability estimates, these improvements do not translate into statistically significant or economically meaningful betting profits. Expected value analysis and rank-based significance testing indicate that higher model-predicted edges do not reliably correspond to higher realized returns. Taken together, these findings support the hypothesis that NBA moneyline betting markets are broadly informationally efficient with respect to publicly available data.

6.2 Future Directions

Several avenues for future research could extend this work. First, incorporating richer data sources, such as player-level injuries, lineup information, or real-time betting market movements, may improve probability estimation and provide a stronger test of market efficiency. Second, evaluating longer out-of-sample periods could increase statistical power and reduce the impact of variance on realized returns.

Additional extensions could explore alternative betting markets, such as point spreads or totals, which may exhibit different efficiency properties. More sophisticated betting strategies, including dynamic stake sizing or line shopping across bookmakers, could also be considered. Another interesting direction would be to incorporate live betting. During the games, the betting odds fluctuate based on the flow of the game. These could also potentially present some profitable opportunities, as there is a possibility that the bookmakers could give odds that are over or under confident in the moment. Finally, alternative modeling approaches, such as hierarchical models or neural networks, may capture additional structure in the data, although their economic value would still need to be evaluated rigorously using out-of-sample expected value analysis.

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A Additional Methodological Details

A.1 ROC Curve and ROC AUC

Let $y_i \in \{0, 1\}$ denote the true outcome for observation i, and let $s_i \in \mathbb{R}$ denote a model score (in this project, the predicted probability of winning). For any threshold t, define the predicted class as $\hat{y}_i(t) = \mathbb{1}\{s_i \geq t\}$.

The *true positive rate* (TPR) and *false positive rate* (FPR) at threshold t are defined as

$$\mathrm{TPR}(t) = \mathbb{P}(s \geq t \mid y = 1), \qquad \mathrm{FPR}(t) = \mathbb{P}(s \geq t \mid y = 0).$$

The ROC curve is the parametric curve $\big(\mathrm{FPR}(t), \mathrm{TPR}(t)\big)$ as t varies over all possible thresholds. The area under the ROC curve (ROC AUC) is defined as

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}(u)\, du,$$

where u denotes the false positive rate.

An equivalent and commonly used interpretation is the probabilistic form:

$$\mathrm{AUC} = \mathbb{P}\left(s^+ > s^-\right) + \frac{1}{2}\mathbb{P}\left(s^+ = s^-\right),$$

where s^+ is the score for a randomly chosen positive instance ($y = 1$) and s^- is the score for a randomly chosen negative instance ($y = 0$). Thus, ROC AUC measures the probability that the model assigns a higher score to a randomly selected win than to a randomly selected loss.

A.2 Log Loss (Cross-Entropy Loss)

Let $\hat{p}_i \in (0, 1)$ denote the model-predicted probability that $y_i = 1$. The log loss (binary cross-entropy) for n observations is defined as

$$\mathrm{LogLoss} = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \log(\hat{p}_i) + (1 - y_i)\log(1 - \hat{p}_i)\right].$$

Log loss evaluates the quality of probabilistic predictions by assigning a large penalty to confident but incorrect predictions (e.g., $\hat{p}_i$ close to 1 when $y_i = 0$). It is minimized when $\hat{p}_i$ equals the true conditional probability.

A.3 Brier Score

The Brier score measures the mean squared error of the predicted probabilities:

$$\mathrm{Brier} = \frac{1}{n}\sum_{i=1}^{n}\left(\hat{p}_i - y_i\right)^2.$$

Lower Brier scores indicate better calibrated and more accurate probabilistic forecasts. Unlike ROC AUC, which depends only on ranking, the Brier score depends on the absolute probability values and therefore reflects probability calibration.

A.4 Expected Value and Realized Return Definitions

This appendix section formalizes the expected value (EV) calculations used to evaluate the economic implications of model predictions.

Let o_i denote the decimal odds offered by the bookmaker for observation i, and let $\hat{p}_i$ denote the model-predicted probability of a win. The expected value of a unit stake bet is

$$\mathrm{EV}_i = \hat{p}_i(o_i - 1) - (1 - \hat{p}_i).$$

Realized return is defined as

$$R_i = \begin{cases} o_i - 1 & \text{if the bet wins } (y_i = 1), \\ -1 & \text{if the bet loses } (y_i = 0). \end{cases}$$

A.5 Spearman Rank Correlation Test

To test whether model-predicted economic value is monotonically related to realized returns, a Spearman rank-order correlation test is used.

Let $\mathrm{rank}(\mathrm{EV}_i)$ and $\mathrm{rank}(R_i)$ denote the ranks of predicted expected values and realized returns, and define d_i as the difference between these ranks for observation i. Spearman's correlation coefficient is

$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2 - 1)}.$$

The null hypothesis is

$$H_0 : \rho = 0,$$

corresponding to no monotonic relationship between predicted edge (as measured by EV) and realized returns.

A.6 Logistic Regression Coefficient Analysis

This subsection provides additional insight into the interpretability of the selected Logistic Regression model. While the main analysis focuses on predictive performance and market efficiency, examining coefficient magnitudes helps illustrate which features contribute most strongly to the model's predicted probabilities.

Figure 5 displays the absolute values of the estimated coefficients from the logistic regression model. Since all input features are standardized, these results provide a meaningful comparison of the relative influence of each variable on the predicted log-odds of a win.

The results indicate that bookmaker-implied probability features dominate the model. In particular, the logit transformation of market probability is by far the most influential predictor, while secondary terms derived from the market (such as squared probabilities and opponent probabilities) contribute modest adjustments. In contrast, contextual variables such as home-court indicators, totals, and spreads exhibit comparatively small coefficients.

This pattern suggests that the predictive signal exploited by the model is largely embedded in the betting market itself. While the model can modify and slightly refine market probabilities, it does not uncover substantial additional information beyond what is already reflected in bookmaker odds. This finding is consistent with the broader evidence of informational efficiency observed in the out-of-sample expected value analysis.

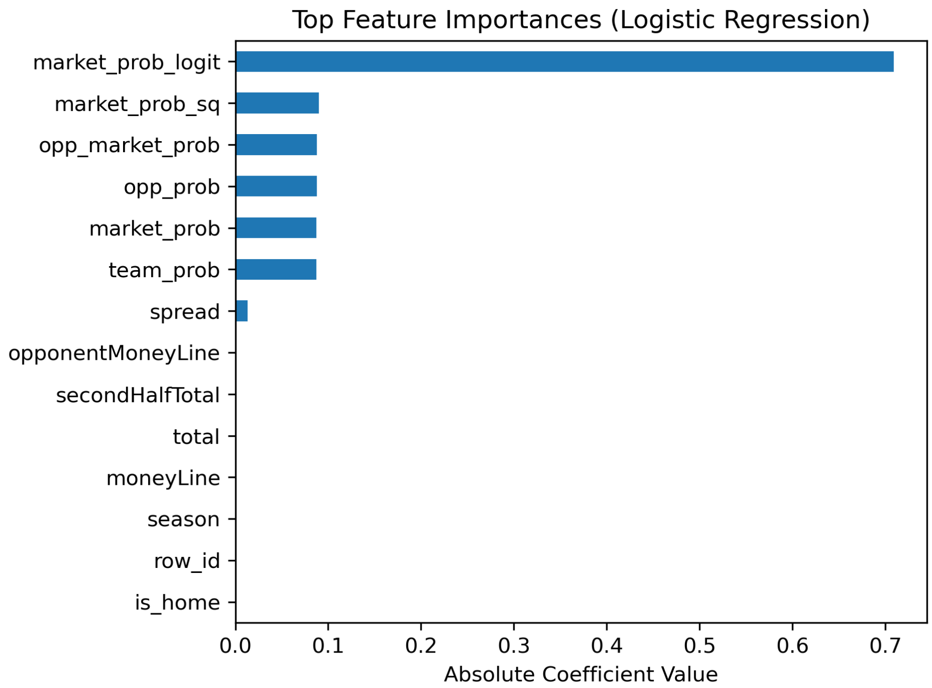

Figure 5: *Standardized coefficient magnitudes for the Logistic Regression model.*

B Code Repository

GitHub Repository: `https://github.com/Markoo99/sports-betting`